Fri 24 Jan 2025

This week we have talked about two concepts:

the stability of a numerical algorithm

and

the conditioning of a mathematical problem

The take away message is that if you have:

a numerically stable algorithm + a well-conditioned problem $\Rightarrow$ will compute accurate results.

then you will compute the exact result of a nearby problem "slightly different"

the solution to the "slightly different" problem will be close to the desired solution

good!

contrapositive

inaccurate result $\Rightarrow$ unstable algorithm or ill-conditioned problem

Reminder on conditioning: $\hat{x} \doteq \bar{x}$

with $\hat{x} = \bar{x}(1+\delta x)$ or expressed as $\dfrac{\hat{x}-\bar{x}}{\bar{x}} = \delta x$

then relative error made when approximating $f(\hat{x})$ by $f(\bar{x})$ is $\left|\dfrac{f(\hat{x})-f(\bar{x})}{f(\bar{x})}\right| \doteq K_f(x)|\delta x|$

magnification factor

error in argument

where $K_f(x) = \left|\dfrac{x f'(x)}{x}\right|$.

The condition number for the mathematical problem.

We saw last time $K_N(x) = 1/2$.

	math	calculator	programming
be careful:	$\ln$	$\ln$	log
	$\log_{10}$	log	log10

<u>example</u> Consider evaluating $f(x) = \ln(x)$.

Observe $\ln(0.999) \doteq -1.0005 \times 10^{-3}$

the result of <u>some</u> calculation

and consider $\ln(0.999(1+0.0001))$

a relative change of 0.01% in the problem

$$= -9.0049 \times 10^{-4}$$

(relate back to previous)

$$= (-1.0005 \times 10^{-3})(1 - \underline{0.09996})$$

a relative change of 9.996% in answer $\doteq 999.6 \times .01\%$

$\therefore$ a small relative change in the problem has lead to a large relative change in the answer $\Rightarrow$ This problem must be ill-conditioned (evaluating $\ln(x)$)

We can quantify the ill-conditioning by using the condition number

$$\kappa_{\ln}(x) = \left| \frac{x \cdot \frac{d}{dx}\ln(x)}{\ln(x)} \right|$$

$$= \frac{|x \cdot 1/x|}{|\ln(x)|} = \frac{1}{|\ln(x)|}$$

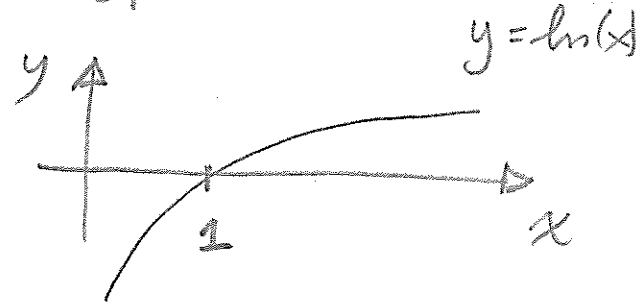
We can use this to explain our observation: when $x = 0.999$

$$\kappa_{\ln}(x) = \left|\frac{1}{\ln(0.999)}\right| \doteq 999.5$$

$\therefore$ the significance of a small error in x can be magnified by ~ 1000 independent of whether stable algorithm used.

Is evaluating $f(x) = \ln(x)$ always ill-conditioned?

We have $K_{\ln}(x) = \dfrac{1}{|\ln(x)|}$



$\ln(x)$ is small in size only for x near 1

$\therefore$ $\ln(x)$ is ill-conditioned only when $x \approx 1$.

$\therefore$ we cannot expect to be able to compute $\ln(x)$ accurately for <u>inexact</u> x near 1. We can expect to be able to compute it accurately for other values of x.

So what should you do when you have x near 1?

If we know $x \approx 1$, let's define a new variable w to be such that

$$x = 1 + w$$

(w is the amount by which x differs from 1)
(set out to determine w accurately, not computing $w = x - 1$)

Then $\ln(x) = \ln(1+w)$.

Now consider the function $g(w) = \ln(1+w)$

$$\kappa_g(w) = \frac{\left| w \cdot \frac{dg}{dw} \right|}{|g(w)|}$$

$$= \left| \frac{w \cdot \frac{1}{1+w}}{\ln(1+w)} \right|.$$

$f(x) = \ln(x)$ is ill-conditioned when $x \to 1$. This corresponds to $w \to 0$

You can show (l'Hôpital) $\lim_{w \to 0} \kappa_g(w) = 1$.

$\therefore$ the function $\ln(1+w)$ is well-conditioned when $w \to 0$!

(This is why math libraries like C libm, python math contain the function $\log1p(a)$.
$\log1p(a) = \ln(1+a)$)

Conclusion: If your "x" could be near 1, compute w instead and compute $\log1p(w)$ when you need $\ln(x)$.